

HELSINKI MALMI, Finland

WMO: 029750

Lat: 60.255N

Lon: 25.043E

Elev: 17

StdP: 101.12

Time Zone: 2.00 (EUE)

Period: 98-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-18.1	-24.1	0.4	-21.7	-20.2	0.6	-17.7	9.6	0.8	9.0	0.9	1.5	350	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.2	18.7	24.9	17.9	23.2	17.0	20.5	24.6	19.2	23.3	18.1	21.8	4.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.7	23.0	17.8	12.8	21.6	16.2	11.6	19.7	58.8	24.4	54.5	23.3	51.1	22.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.6	7.7	6.9	-23.3	28.7	4.8	1.5	-26.8	29.7	-29.6	30.6	-32.3	31.5	-35.8	32.5		
			-23.6	21.2	4.7	1.2	-27.0	22.1	-29.7	22.8	-32.4	23.4	-35.9	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-4.7	-4.5	-1.8	4.3	10.3	14.2	17.9	16.1	11.8	6.0	1.9	-1.5
	DBStd	8.88	6.35	6.03	4.79	3.19	3.77	3.00	2.72	2.66	2.96	3.85	4.50	5.66
	HDD10.0	2192	457	406	366	173	42	3	0	1	16	131	242	357
	HDD18.3	4585	715	639	624	421	252	128	42	78	196	383	492	616
	CDD10.0	692	0	0	0	2	49	129	244	190	70	7	0	0
	CDD18.3	43	0	0	0	0	1	5	28	9	0	0	0	0
	CDH23.3	320	0	0	0	0	9	46	212	54	0	0	0	0
	CDH26.7	39	0	0	0	0	0	4	33	3	0	0	0	0
Wind	WSAvg	3.0	3.2	3.0	3.0	3.0	3.0	3.0	2.8	2.7	2.8	3.0	3.3	3.6
Precipitation	PrecAvg	662	47	33	33	36	36	52	68	79	68	76	72	59
	PrecMax	903	105	112	78	101	86	124	201	192	150	193	216	168
	PrecMin	422	8	3	3	3	3	12	4	9	12	16	5	8
	PrecStd	120	25	22	16	21	19	29	43	39	35	42	39	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	5.9	11.2	17.8	24.8	26.9	29.1	27.1	21.0	14.9	10.8	8.9
		MCWB	5.2	3.8	6.2	10.1	17.1	17.3	20.2	19.1	15.8	12.3	9.5	8.1
	2%	DB	4.2	3.8	8.0	14.9	22.1	24.8	27.2	25.1	19.2	13.2	9.2	7.1
		MCWB	3.7	3.1	4.2	8.7	14.5	16.8	19.9	18.2	14.9	11.6	8.5	6.4
	5%	DB	3.2	2.9	5.9	12.8	20.1	22.8	25.9	23.8	18.1	12.8	8.8	6.0
		MCWB	2.7	2.3	3.1	7.5	13.6	15.7	18.7	17.5	14.4	11.3	8.1	5.3
	10%	DB	2.8	2.0	4.0	10.8	17.9	20.9	24.0	22.1	17.0	11.8	7.8	5.0
		MCWB	2.3	1.5	2.5	6.1	12.5	15.1	18.0	16.8	14.1	10.5	7.3	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	3.9	6.7	10.5	18.2	19.7	21.8	21.6	16.8	13.2	9.7	8.0
		MCDB	5.3	4.4	10.1	16.0	23.0	23.1	26.6	25.0	19.3	14.1	10.1	8.5
	2%	WB	3.7	2.9	4.7	9.3	16.2	17.7	21.0	19.5	15.9	12.5	8.6	6.4
		MCDB	4.1	3.3	6.8	13.7	19.9	22.5	25.5	23.1	18.1	13.3	9.0	6.8
	5%	WB	2.8	2.1	3.4	7.9	14.8	16.5	20.0	18.5	15.2	11.6	7.8	5.2
		MCDB	3.2	2.5	5.0	11.4	18.7	21.2	24.3	22.1	17.3	12.6	8.2	5.7
	10%	WB	1.9	1.3	2.4	6.5	12.8	15.3	18.7	17.6	14.5	10.5	6.9	4.1
		MCDB	2.2	1.7	3.5	9.8	17.3	19.6	22.8	20.8	16.5	11.5	7.3	4.7
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.3	7.4	8.3	9.5	8.9	8.8	8.8	7.7	6.4	4.3	4.8
		MCDBR	3.9	4.4	9.3	12.6	12.8	12.3	11.5	11.9	9.3	6.9	4.0	3.9
		MCWBR	3.9	4.0	6.2	7.3	6.5	5.8	5.3	6.0	6.0	5.3	4.0	3.8
	5% WB	MCDBR	3.9	3.6	7.7	10.9	10.1	10.5	10.0	9.3	7.8	5.6	3.9	3.6
		MCWBR	3.9	3.3	5.5	6.8	6.0	5.7	5.1	5.3	5.6	4.8	4.0	3.6
Clear-Sky Solar Irradiance	taub	0.278	0.312	0.334	0.359	0.358	0.355	0.377	0.370	0.343	0.333	0.305	N/A	
	taud	2.138	2.191	2.248	2.348	2.417	2.475	2.436	2.453	2.518	2.394	2.214	N/A	
	Ebn at Noon	471	669	785	838	869	876	845	821	786	653	435	N/A	
	Edh at Noon	43	72	94	102	103	99	101	92	73	60	40	N/A	
All-Sky Solar Radiation	RadAvg	0.25	0.82	2.31	3.84	5.23	5.44	5.25	4.06	2.48	1.08	0.30	0.15	
	RadStd	0.04	0.16	0.30	0.40	0.56	0.43	0.48	0.42	0.32	0.17	0.06	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	-142	N/A	N/A	N/A

Nomenclature: See separate page